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Multispectral imaging approach in the diagnosis of cutaneous melanoma: potentiality and limits

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Abstract. In an attempt to overcome the subjectiveness of clinical observation in the diagnosis of cutaneous melanoma, a computerized method is proposed. Reflectance images of 237 pigmented lesions (67 melanomas and 170 non-melanomas) were analysed using a telespectrophotometric technique. This device consists of a CCD camera with 17 interference filters. Images were acquired at selected wavelengths, from 420 to 1040 nm. Morphological and reflectance related parameters were extracted from the wavelength-dependent images of the lesions. The most significant features in the comparison between benign and malignant lesions were: lesion dimension ($P < 10^{-8}$ at 578 nm); mean value ($P < 10^{-7}$ at 940 nm) and standard deviation ($P < 10^{-4}$ at 904 nm) of lesion reflectance; lesion roundness ($P < 10^{-5}$ at 461 nm); and border irregularity ($P < 10^{-4}$ at 461 nm). Based on these parameters, a discriminant function between the two populations of lesions (naevi and melanomas) was obtained. By using the results of the analysis of the recruited lesions as 'training data', discriminant functions enabled the assignment of a score, or a 'risk probability', to each studied lesion. By imposing a sensitivity of 80% (a figure that mimics the diagnostic capability of an experienced clinician), entering or not entering the lesion dimension as input data in the discriminant analysis led to a specificity of 51% or 46% respectively.

The high number of false-positive cases, which is a consequence of the selection criteria of the lesions, is, at present, the major limitation of the current technique. Nevertheless, our results suggest that an imaging-based computer-assisted device could be capable of discriminating malignant lesions mainly by evaluation of reflectance, especially in the infrared region, and shape properties. The dimension of a lesion should not be essential in the diagnosis of melanoma and, in our opinion, small melanomas should be recognized by a computer system as well as they are on clinical grounds.

1. Introduction

The dramatically increasing incidence rate of cutaneous melanoma (CM) has prompted primary care physicians and specialists to learn to recognize this condition in its early, curable phase. Clinical diagnosis is based on observation of the colour and shape features of a skin lesion. Nevertheless, visual diagnosis of pigmented lesions is often a difficult diagnostic problem, even among experienced clinicians. As with any clinical diagnosis, the evaluation of a pigmented lesion is to a large extent subjective and strictly related to the experience of the clinician.

In recent years several techniques have been proposed to overcome this problem, which, based on computer image analysis, attempt to provide the required quantitative measurements in an objective and reproducible fashion (Green *et al* 1994, Binder *et al* 1998). Since 1994, a

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telespectrophotometric technique, based on the analysis of a sequence of multispectral images of a lesion, has been under development at the Istituto Nazionale Tumori in Milan to investigate the colour and shape features of a skin lesion. Feasibility studies and preliminary results of this technique have been previously published (Bono *et al* 1996).

In the present pilot study we investigate whether a computerized system, able to extract lesion features when applied to multispectral images, can provide a correct classification of a series of suspicious pigmented lesions scheduled to undergo surgery.

2. Materials and methods

2.1. Study population

Between January 1995 and December 1997, 226 patients (82 males, 144 females) with 237 cutaneous pigmented lesions (including 67 cutaneous melanomas) were enrolled in this study at the Istituto Nazionale Tumori, Milan. The average age of the patients was 40 years (range 8–95). One hundred and forty five lesions (61%) were located on the trunk, 84 (35%) on the limbs and 8 (4%) on head or neck. The maximum diameter of the lesions ranged from 4 to 27 mm, with a mean value of 12.7 mm, and from 2.5 to 26 mm, with a mean value of 8.8 mm, for cutaneous melanomas and all the other lesions respectively. We excluded from the study lesions which were clearly appearing as thick and large melanomas or which were awkwardly situated (e.g. placed in the interdigital spaces, on ears, nose, eyelids), as well as lesions of the scalp, due to interference from the hair in imaging processing. Excluding the lesions that did not need image analysis as they were clinically evaluated as apparent malignant or benign naevi, the percentage of suspicious lesions for which the acquisition of images was not possible amounted to no more than 5% of the total cases. The slides were evaluated according to the widely accepted criteria for histopathological diagnoses of various pigmented lesions (Elder and Murphy 1991). Table 1 shows the histological nature of cutaneous melanomas (CM) and non-melanoma lesions (NM). Among the CM, the median invasion thickness was 0.68 mm, ranging from 0.18 to 3.01 mm, 40 were thin lesions, 8 *in situ* and 32 with tumour thickness $<$0.76 mm (Balch *et al* 1992). Six CM were less than 6 mm in diameter.

Table 1. Histological diagnosis of 237 pigmented lesions.

Diagnosis	Number of cases (%)
Cutaneous melanoma (CM)	67 (28.3%)
Invasive	59 (24.9%)
In situ	8 (3.4%)
Non-melanoma (NM)	170 (71.7%)
Compound nevus	80 (33.8%)
Dysplastic nevus	30 (12.7%)
Junctional nevus	21 (8.9%)
Dermal nevus	8 (3.4%)
Lentigo simplex	7 (3.0%)
Spindle-cell nevus	5 (2.1%)
Spitz nevus	4 (1.7%)
Basal cell carcinoma	3 (1.3%)
Blue nevus	5 (2.1%)
Seborrheic keratosis	2 (0.8%)
Other	5 (2.1%)
Total	237 (100%)

2.2. *Imaging system*

The image acquisition system consists of a Sony CCD camera (604×588 sensitive elements), a set of 17 interference filters, an IBM-compatible personal computer and an illumination system composed of two halogen lamps (2×100 W) and two lamps (2×150 W) with emission in the infrared region. A rotating wheel mounted inside the camera holds the filters and is placed between the 50 mm camera lens and the CCD detector. Each filter is automatically positioned by software control. The filter-to-filter wavelength interval is about 40 nm in the 420–1040 nm range. Filters have a peak transmittance that ranges from 50 to 80%. Full-width-at-half-maximum of the transmission peak varies from 20 to 35 nm. The halogen and infrared lamps, assembled on a stand that also supports the CCD video camera, have a diffuser to ensure evenly distributed illumination on the skin surface of interest (an area of 4×5 cm^2, including both lesion and normal skin). A frame grabber installed in the computer allows to capture a lesion's reflectance image (320×200 pixels) to be captured at each of the 17 selected wavelengths, in 256 grey levels and with a spatial resolution of 3.5 pixel/mm. The acquired images were stored in the PC for off-line data handling (Marchesini *et al* 1995).

2.3. *Digital images analysis*

To convert the image raw numbers into absolute reflectance, a set of four reflectance standards (i.e. 10%, 20%, 40% and 60%), specially manufactured, calibrated and packaged in a 1.2×3.8 cm^2 frame (Labsphere, Sutton, NH), were placed near the lesion. Each acquired reflectance image included a graduated frame as reference for spatial calibration.

The segmentation of the image, to contour the lesion from the normal skin background, was carried out by means of a thresholding technique (Hall *et al* 1995) and automatically performed by the image analysis software. Manual correction of the outline was not allowed. Only the pixels within a grey-scale range between 0 and 80% of the mean reflectance value of the normal skin surrounding the lesion were selected. The segmentation was essential for the evaluation of a set of parameters (lesion descriptors) which quantified the lesion features. A software procedure was developed to fully automate the whole process of calibration, segmentation and evaluation of the lesion descriptors.

For each spectral image, five parameters (lesion descriptors) related to colour and shape of the imaged lesion were evaluated (table 2). The choice of the descriptors was made by taking into account some of the lesion features that alert clinicians and which can be related, to some extent, to the ABCD rule (Friedman and Rigel 1985). The first two descriptors are related to the colour features of the lesion. Mean reflectance (MR) measures the ability of a lesion to reflect/diffuse the incident light and is related to the colour of the lesion; variegation index (VI) is an indicator of how evenly light is reflected/diffused and is related to the pigment distribution inside the lesion. The other parameters describe size and shape of a mole. In particular, roundness (R) is an estimate of how a lesion contour resembles a circle ($R = 1$ for a perfect circle), whereas border irregularity (BI) is an indicator of the lesion border regularity.

The variables related to lesion shape features (i.e. area A, R, BI) were determined only at wavelengths in the visible range (420–623 nm), where the contrast between skin and lesion border was high enough to allow the system to recognize the contour of all the studied lesions. MR and VI were evaluated in the visible and near-infrared wavelengths (420–1020 nm). When, especially in the near-infrared, the lesion was no longer clearly detectable, evaluation of MR and VI was based upon the contour determined at the last wavelength where the lesion could still be imaged (Marchesini *et al* 1995). The images obtained from the 1040 nm filter were not included in the study because they appeared blurred, probably due to the drop in sensitivity of

Table 2. Numerical parameters extracted from image analysis to characterize colour (MR, VI) and shape (A, R, BI) features of cutaneous lesions.

	Definition
Mean Reflectance (MR)	Mean value of the reflectance of the segmented lesion
Variegation Index (VI)	Standard deviation of the reflectance of the segmented lesion
Area (A)	Area of the segmented lesion
Roundness (R)	$\frac{4\pi \times \text{area}}{(\text{lesion perimeter})^2}$
Border Irregularity (BI)	$\frac{\text{perimeter}_C}{\text{lesion perimeter}}$ (where 'perimeter$_C$' is the perimeter of the convex hull enclosing the lesion)

the CCD camera at this wavelength, thus compromising a reliable measure of the reflectance of the lesion as well as that of the normal skin.

2.4. Statistical analysis

Following image processing and evaluation of lesion descriptors, two statistical analyses were performed on the collected data. Since we were interested in evaluating the different behaviour of malignant naevi with respect to benign moles, a Student's t-test was performed first on the mean values of each lesion descriptor. The following step was directed to establish a discriminating tool capable of classifying a lesion to the group of CM or to the group of NM, according to its colour and morphological features. The 237 pigmented lesions enrolled in this study were used to obtain a 'training-data set' for a stepwise discriminant analysis. The stepwise procedure in linear discriminant analysis is an established method for selecting useful features (variables) for a classification task. A detailed description of this procedure can be found in the literature (Kleinbaum *et al* 1988). Briefly, the stepwise analysis operates by including one variable at each step in a discriminating function (DF) until no further gain in the separation between groups results from insertion of the other variables. The effect of the variable on the separation of the two groups is analysed using the Wilks lambda criterion (i.e. minimization of the ratio of the within-groups variance to between-groups variance). The variables entering as input data in the discriminant analysis were chosen by selecting, for each lesion descriptor, the wavelength-dependent value that showed the minimal P-value of the Student's t-test when CM were compared with NM. Concomitantly, the lesion descriptors were also tested for the subgroup of dysplastic naevi in comparison with the other pigmented lesions. Student's t-test and discriminant analysis were carried out using a commercially available software (STATISTICA/W, StatSoft, Tulsa, OK, USA).

3. Results

3.1. Comparison between benign and malignant nevi

Figure 1 reports a typical reflectance image of a cutaneous pigmented lesion and an example of images acquired at 461, 623 and 940 nm of a CM and a benign lesion. The lesion appearance is clearly wavelength dependent, and at 940 nm the benign naevus shows a reflectance close to that of the surrounding normal skin.

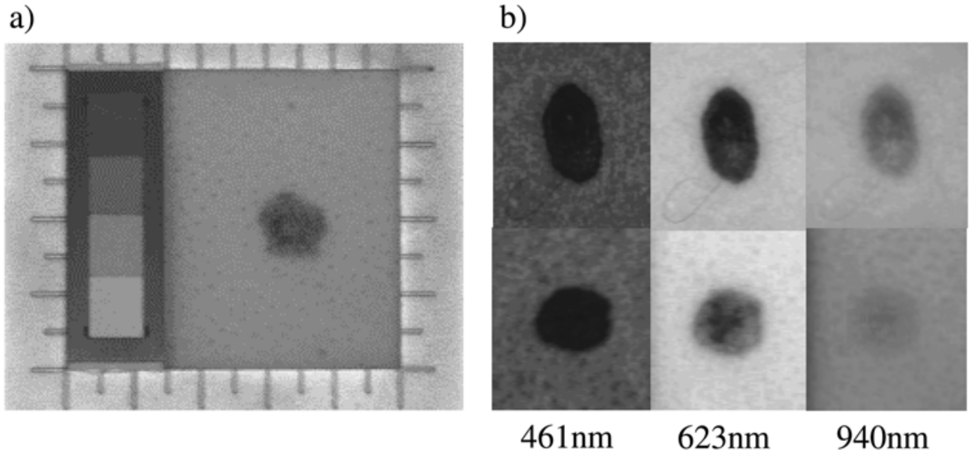

Figure 1. (*a*) Reflectance image of a cutaneous pigmented lesion. The image was acquired at a wavelength of 538 nm. Four reflectance standards and a graduated frame are placed near the lesion. (*b*) Typical images at selected wavelengths of a melanoma (top) and a benign lesion (bottom). At 940 nm the reflectance of the benign naevus is close to that of the surrounding skin, whereas pigmentation of the melanoma is still detectable.

Figure 2 reports the spectral behaviour of the lesion descriptors according to the three groups of melanomas, dysplastic naevi and other pigmented lesions. CM showed lower reflectance over the whole range of wavelengths investigated when compared with NM. This finding agrees with the common clinical impression that melanomas are usually very dark pigmented lesions. Interestingly, both colour parameters (MR and VI) become more significantly different when CM are compared with NM in the red and, especially, in the infrared region. The morphological descriptors (R, BI) indicate that CM are less rounded and have a less regular contour with respect to NM and, in contrast to MR and VI, the significance level increases at shorter wavelengths. As regards the size of the lesions, CM appear larger than NM ($P < 10^{-7}$). Finally, comparison between lesion descriptors of dysplastic naevi and other pigmented lesions did not show any significant difference.

3.2. Discriminant analysis and classification

From the results of the Student's t-test we extracted a set of variables to perform discriminant analysis (table 3). Since dysplastic naevi did not give a significantly different result from the benign nevi, the discriminant analysis was performed between CM and NM. By using as input data either MR, VI, A, R, BI or MR, VI, R, BI, i.e. including or not including the lesion area, two discriminant analyses were carried out, which resulted in the following discriminant functions (DF_1 and DF_2)

$$DF_1 = -4.07 + 8.15(MR_{940}) - 0.84(A_{578}) + 1.89(R_{461})$$
$$DF_2 = -6.73 + 10.19(MR_{940}) + 3.70(R_{461}).$$

In both cases, variables VI_{904} and BI_{461} have been neglected by the discriminant functions because they were no longer significant in discriminating the two groups of lesions when inserted in the stepwise analysis together with the other variables.

Figure 3 shows the distribution of the scores assigned to each case by the two discriminant functions. For representative purposes, lesions have been grouped in unit score bins.

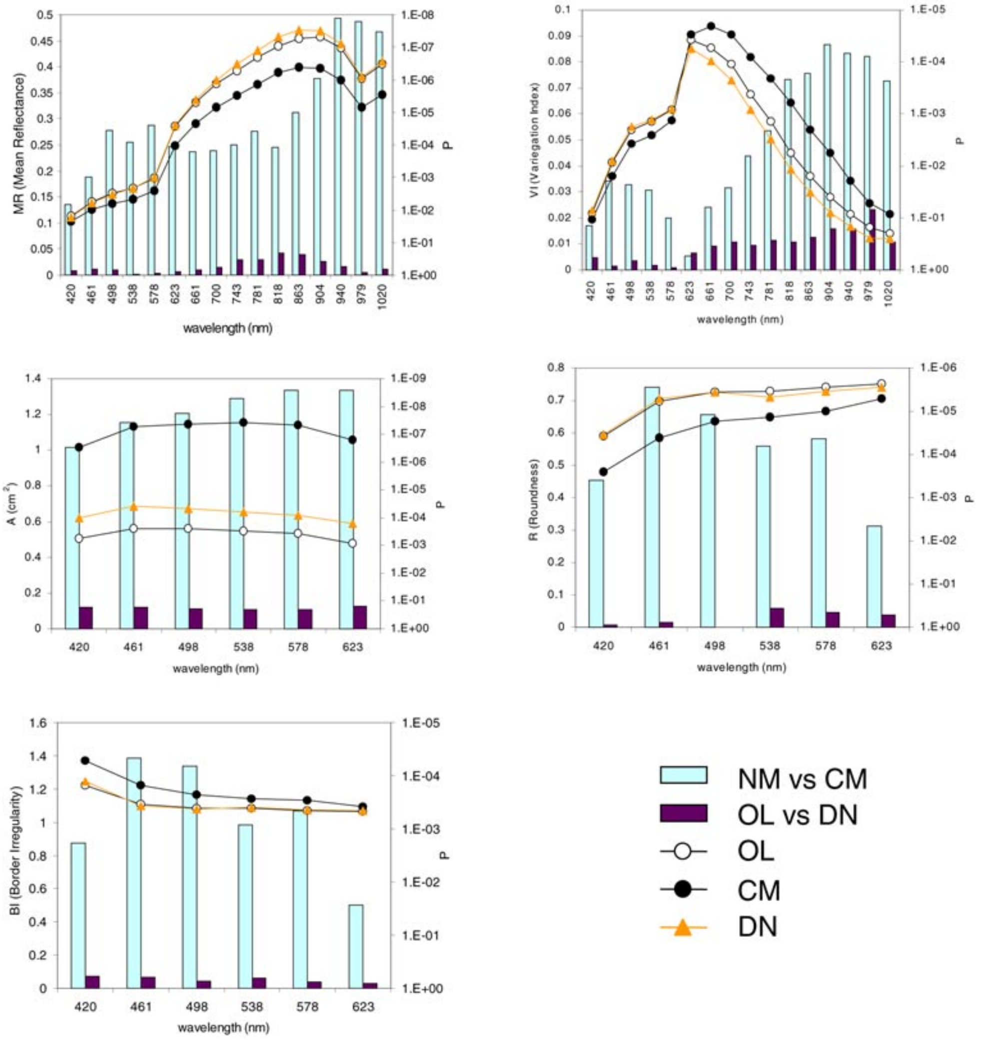

Figure 2. Lesion descriptors. Solid symbols: mean value of melanoma (CM), dysplastic naevi (DN) and other pigmented lesions (OL). Bars: P value of Student *t*-test, when comparing NM with CM and OL with DN.

CM lesions have been separated into three groups: *in situ*, with thickness lower than 0.76 mm and with thickness greater than 0.76 mm (Balch *et al* 1992). In order to evaluate the capability of our imaging device to classify small melanomas, the score of CM lesions with maximum diameter less than 6 mm have also been represented. From the results of the discriminant analysis, a linear correlation between the sequence BN $\rightarrow$ CM *in situ* $\rightarrow$ CM < 0.76 mm $\rightarrow$ CM $\geqslant 0.76$ mm and the mean value of the scores of each group has been observed ($R = -0.96$ and -0.88 for DF_1 and DF_2 respectively).

3.3. Classification of the cases

The classification of the cases depends on the threshold applied to separate the output values of the two DFS. An overview of the system discriminating power is shown in the receiver

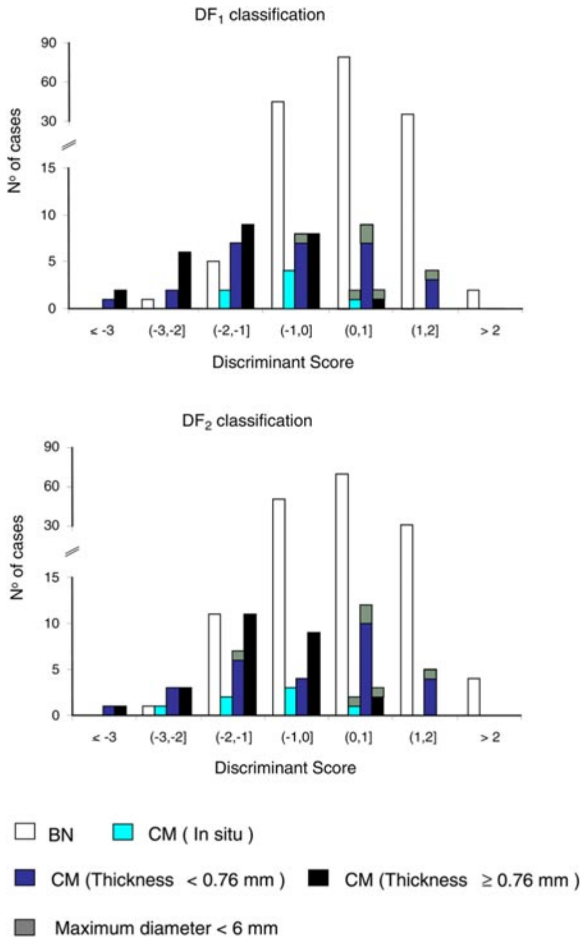

Figure 3. Distribution of the scores, according to histology, evaluated by the discriminant functions.

Table 3. Variables entering as input data in the discriminant analysis. The number next to the variable's name stands for the corresponding wavelength.

Variable	NM (mean value ± SD)	CM (mean value ± SD)	t-value	P
MR_{940}	0.44 ± 0.070	0.37 ± 0.084	5.89	1.3×10^{-8}
VI_{904}	0.028 ± 0.027	0.045 ± 0.032	−4.15	4.7×10^{-5}
A_{578}	0.55 ± 0.41	1.14 ± 1.06	−6.19	2.3×10^{-9}
R_{461}	0.70 ± 0.14	0.59 ± 0.20	4.81	3×10^{-6}
BI_{461}	1.11 ± 0.157	1.22 ± 0.25	−4.14	4.7×10^{-5}

operating characteristic (ROC) data reported in figure 4. The ROC is a curve showing the various trade-offs existing between proportions of true-positive and false-positive responses, as the decision criterion is systematically varied, i.e. for a given capacity to discriminate between positive and negative cases (Thompson and Zucchini 1989). From a diagnostic test making a complete discrimination between two distributions, an ROC curve moving towards the left and top boundary of the graph will result; on the contrary, with experimental values unable

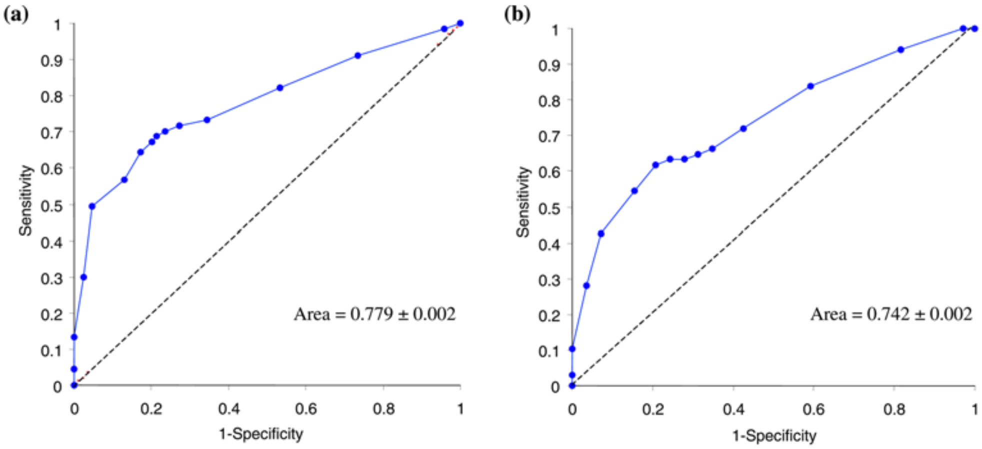

Figure 4. ROC data (solid symbols) for (*a*) DF_1 and (*b*) DF_2 discriminant functions.

to make discrimination, the ROC curve approaches the diagonal line. Therefore, the most common quantitative index describing a ROC curve is the area under it. This parameter can assume values ranging from a minimum of 0.5, representing chance behaviour, to a maximum of 1.0, representing perfect discrimination. In this study the areas beneath the ROC curves have been obtained by the trapezoidal rule which systematically underestimates this parameter (Hanley and McNeil 1982).

Different criteria can be applied in choosing a cut-off that results in optimal classification of the examined lesions. By imposing a threshold of 0.35 and 0.32 on the output value (i.e. discriminant score) of DF_1 and DF_2 respectively, discrimination between CM and NM resulted in a sensitivity of 80%. This figure can be roughly considered as an indication of the diagnostic capability of an experienced clinician (Grin *et al* 1990, Wolf *et al* 1998). As regards the CM small lesions (i.e. maximum diameter less than 6 mm), one and two cases out of six were correctly classified as CM by DF_1 and DF_2 respectively. Table 4 reports the resulting values of specificity, negative predictive value (NPV) and positive predictive value (PPV) which have been derived from DF_1 and DF_2, as well as the corresponding figures derived from the clinical judgement.

4. Discussion

Efforts to educate primary care physicians and the general public about the distinctive clinical features of melanoma are being developed in an effort to decrease the mortality of the disease. Indeed, early detection and prompt excision of melanoma give the patient the best chance of cure (Balch *et al* 1992). Since clinical diagnostic accuracy remains a challenge, a number of technological tools have been developed to aid in the diagnosis of pigmented skin lesions. Epiluminescence microscopy is the most developed of these techniques (Pehamberger *et al* 1993) and imaging-based analysis systems are being developed by several groups (Green *et al* 1994, Cascinelli *et al* 1987, Binder *et al* 1998).

Imaging systems usually work in the visible region of the electromagnetic spectrum and provide evaluation of naevi dimension as well as the lesion's shape and contour. This study confirms that melanomas differ from naevi in border and shape features, especially at the shortest wavelengths (green and blue) in the visible. As regards the colour properties of skin

Table 4. Comparison between the classification resulting from the discriminant analysis and clinical diagnosis.

		No of corrected diagnoses		
Histology	No	Clinical	DF_1	DF_2
Melanoma (CM)	67	61	54	54
Non-melanoma (NM)	170	131	87	79
Total	237			
Sensitivity (%)±SD	91 ± 3	80 ± 5	80 ± 5	
Specificity (%)±SD	77 ± 3	51 ± 3	46 ± 4	
PPV (%)	61	39	37	
NPV (%)	96	87	86	

lesions, our study has shown that melanomas have a lower reflectance and their pigmentation is less uniform with respect to benign naevi. These differences are greatly enhanced in the red and infrared regions, thus confirming, as previously suggested, that colour represents the strongest feature useful for diagnosis of melanoma (Bono *et al* 1999a).

In an attempt to explain the reasons for the observed effects some hypotheses can be suggested. Melanin present in melanomas might have optical properties different from those of melanin in benign naevi, and it might aggregate differently in the two populations (Marshall 1976). In addition, melanosomes show a different fine structure and a range of abnormalities (Takahashi *et al* 1985) and could play a significant role in modifying the reflectance spectrum by acting as added scattering and absorbing centres to the normal pigmentation of benign naevi. Another source of added absorbers might be related to the different vascular structure and the abnormal blood flow that melanomas present with respect to that of benign naevi (Srivastava *et al* 1986). Finally, highly aggregated cells (e.g. lymphocytes, fibroblast and macrophages) are usually present in melanoma. The presence of such cells in the skin layers could greatly affect scattering and/or absorption of light, which in turn affects the light-reflectance pattern of the tissue. It all could reasonably indicate some of the biophysical features involved in the difference in reflectance and its variation with wavelength between CM and benign pigmented lesions. To fully elucidate the actual reasons for the observed difference in the morphological parameters, a quantitative evaluation of absorption and scattering properties of malignant and benign lesions should be carried out. However, measurements on *ex vivo* lesion samples are not allowed before performing histopathological diagnosis, for ethical reasons, and after histology the specimen is no longer suitable for evaluating optical characteristics.

The variation in the significance level of the lesion area with wavelength is mainly due to the fact that as the wavelength increases images of benign lesions fade with a trend faster than that shown by CM (figure 1). As a consequence, the difference between areas, measured after segmentation, of CM and benign naevi increases. Conversely, since roundness and border irregularity are related to the spatial resolution of the lesion outline, the less faded the images the more the details of lesion outline which can be evaluated as wavelength decreases. The reason for the drop in the significance level in passing from 461 to 420 nm remains unclear. A possible explanation might be related to the lower penetration depth in tissue of the 420 nm wavelength with respect to longer ones.

Our imaging technique allows a characterization of a cutaneous lesion according to five numerical descriptors, related to the shape and reflectance features of the lesion, which, when used as input data in a multivariate statistical analysis enabled us to assess a discriminant function to assign a classification score to the imaged lesion. Results found in the literature report values of clinical sensitivity between 67 and 91% (Grin *et al* 1990, Miller and Ackermann 1992, Witheiler and Cockerell 1992, Morton and MacKie 1998). Imposing on our device a sensitivity of 80%, considered close to the diagnostic capability of an experienced clinician, a relatively poor specificity resulted (51% and 46% in the two different analyses).

Comparison with data reported by other authors, who used automated methods, suggests that two features, number and characteristics of the studied cases, play the main role in the performance of computerized image analysis for melanoma detection. Green *et al* (1994) collected 204 lesions and investigated 17 variables obtained from an RGB-image computer analysis of 164 cases, including 18 melanomas. Eightynine per cent of cases were correctly classified, area and perimeter of the lesion being the main determinants in discriminating melanomas from benign lesions. Moreover, very few data on lesion dimension and invasion thickness were reported. It is worth mentioning that a preliminary study performed with our technique on 51 lesions, including 18 melanomas with a median invasion thickness of 0.845 mm, resulted in a sensitivity and specificity of 89% and 88% respectively (Tomatis *et al* 1998). Binder *et al* (1998) investigated, using epiluminescence microscopy and an artificial neural network, 16 morphometric parameters of 120 lesions, including 39 melanomas with a median invasion thickness of 0.72 mm. Five per cent of the lesions were withdrawn due to incorrect segmentation of the automatic image analysis. The sensitivity and specificity of the computerized system were dependent on the selection of the lesions and varied from 93% to 38% and from 84% to 50% respectively. Our data and these findings suggest that the diagnostic capability of computerized image analysis is greatly related to the characteristics of the recruited lesions, and that any claimed accuracy of a computer-assisted technique for the diagnosis of melanoma should be assessed on the basis of well defined, even if at present not yet standardized, features of the lesions enrolled in the study.

The high number of false-positive cases, are, in our opinion, a consequence of the selection criteria of lesions (table 1) enrolled in this study. In fact, all lesions were judged suspicious or at least doubtful with the necessity of histological verification. The resulting high value of sensitivity in clinical diagnosis of melanoma (91%) was due to the longstanding experience of the clinicians of our group. A similar high value of sensitivity was achieved when diagnosis was made only by dermatologists who had over 10 years of clinical experience (Morton and MacKie 1998).

The size of a lesion is one of the main features that alerts clinicians, and is the most significant parameter when CM is compared with NM by statistical test, as shown in this paper and other similar studies (Grin *et al* 1990, Cascinelli *et al* 1992). Moreover, it is interesting to note that the results obtained by DF_2 (i.e. area not included in the discriminant function) are similar to those obtained by DF_1. This finding suggests that an imaging-based computer-assisted device could be capable of discriminating malignant lesions mainly by evaluation of reflectance and shape properties. The dimension of a lesion might not be essential in the diagnosis of melanoma and, in our opinion, small melanomas should be recognized by a computer system as well as they are on clinical grounds (Bono *et al* 1999b).

However, when dealing with imaging-based computer-assisted system for detection of melanoma special consideration should be devoted to the characteristics of the pigmented lesions under study. Usually, the published papers on this topic do not fully report information on the histology of the malignant lesions analysed (e.g. thickness, Clark's level, dimension), nor the reason to include or not in the study 'common' pigmented lesions which will eventually be compared with melanomas in the subsequent discrimination procedure. As a consequence, any claimed sensitivity and specificity, especially if greater than that of an experienced clinician, should be judged very carefully. In addition, the lesion dimension seems to play the main role in discriminating malignant from benign moles and, if so, a cheap ruler could be used rather than expensive instrumentation. In our opinion, it is hard to assess whether there will be a future for computer-assisted diagnosis of melanoma until evidence is reported that melanoma is recognized not only on the basis of the lesion dimension.

By imposing a sensitivity of 80%, DF_2 resulted in a slightly lower specificity than that obtained by DF_1, but a decrease in misclassification of small (<6 mm) melanomas was obtained. The very small number of cases does not allow us to be sure of the reasons for such a difference nor to what extent our imaging system will permit us to recognize early melanomas. Since roundness is one of the parameters that the discriminant functions utilize for discrimination, work is in progress to improve the spatial resolution of the acquired images in an attempt to emphasize the role of that lesion descriptor.

A study is ongoing to make the telespectrophotometric system a practical diagnostic tool for routine use. The images used in this study have been used to achieve a training data set to determine the diagnostic parameters and make the system affordable and easy to use, before proving it on lesions of unknown histology. The system is anyway intended for 'screening and second opinion', and not to replace a clinician's skill but to complement it with non-invasive diagnosis.

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